Panskura Banamali College (Autonomous)

B.Sc.(H) 1st Semester Examination 2022 (Online mode)

Subject : Computer Science Paper: CC2T

Paper Name : Computer System Architecture

F.M. : 40 Time : 2 Hours

Answer any 04 questions : 4×10=40

1. a) What is logic gate ? What is truth table ? What is the purpose of universal logic gate?
b) . Draw a NAND logic diagram that implements the following function :
F(A,B,C,D) = Σ(1,3,5,8,9,11,15) + d (2,13) 2+2+1+5=10

2. a) Implement the Boolean function f(a,b,c,d)= Σ(0,1,2,3,5,7,9,14) using multiplexer.
b) What is bus? Describe the bus architecture of a Micro-computer System . 5+2+3=10

3. a) What are the difference between decoder and demultiplexer ? What is active high and active low signal?
2+3=5
b) What do you mean by priority encoder ? What are the difference between synchronous and asynchronous sequential circuits ? 2+3=5

4. a) Realize two input AND gate using NOR gates.
b) Consider the Boolean function f(a,b,c,d)= Σ(0,1,4,5,7,8,14). Minimize the Boolean function using Karnaugh Map.
c) Show that ABC+B+BD+ABD+AC=B+C 3+4+3=10

5. a) What is encoder ? Explain how the decimal digits 0 to 9 can be encoded as BCD using basic gates. 2+3=5
b) Using JK flip-flops draw a block diagram of a 4-bit asynchronous counter. How many clock pulses does it count in one cycle of its operation ? What is literal ? 3+1+1=5

6. Write short notes : (any four) 4×2.5=10
   (i) DMA data transfer
   (ii) Von Neumann architecture
   (iii) Pipelining
   (iv) Shift Register.
   (v) Cache memory

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